

# Arcanum: Detecting and Evaluating the Privacy Risks of Browser Extensions on Web Pages and Web Content

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# Arcanum: Detecting and Evaluating the Privacy Risks of Browser Extensions on Web Pages and Web Content

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Modern web browsers support rich extension ecosystems that provide users with customized and flexible browsing experiences. Unfortunately, the flexibility of extensions also introduces the potential for abuse, as an extension with sufficient permissions can access and surreptitiously leak sensitive and private browsing data to the extension's authors or third parties. Prior work has explored such extension behavior, but has been limited largely to meta-data about browsing rather than the contents of web pages, and is also based on older versions of browsers, web standards, and APIs, precluding its use for analysis in a modern setting.

In this work, we develop Arcanum, a dynamic taint tracking system for modern Chrome extensions designed to monitor the flow of user content from web pages. Arcanum defines a variety of taint sources and sinks, allowing researchers to taint specific parts of pages at runtime via JavaScript, and works on modern extension APIs, JavaScript APIs, and versions of Chromium. We deploy Arcanum to test all functional extensions currently in the Chrome Web Store for the automated exfiltration of user data across seven sensitive websites: Amazon, Facebook, Gmail, Instagram, LinkedIn, Outlook, and PayPal. We observe significant privacy risks across thousands of extensions, including hundreds of extensions automatically extracting user content from within

web pages, impacting millions of users. Our findings demonstrate the importance of user content within web pages, and the need for stricter privacy controls on extensions.

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